Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr.S60 (N&S), Elite, IAS & LIIT | JEE-MAIN- Shift-2 | Date: 30-03-2026

Time: 03.00Pm to 06.00Pm | GTM-33 | Max. Marks: 300

KEY SHEET

MATHEMATICS

1	**2**	2	**3**	3	**1**	4	**4**	5	**3**
6	**4**	7	**2**	8	**3**	9	**3**	10	**2**
11	**4**	12	**1**	13	**2**	14	**3**	15	**4**
16	**2**	17	**2**	18	**3**	19	**1**	20	**4**
21	**50**	22	**2**	23	**24**	24	**1**	25	**3**

PHYSICS

26	**1**	27	**4**	28	**1**	29	**1**	30	**3**
31	**1**	32	**3**	33	**3**	34	**2**	35	**3**
36	**1**	37	**1**	38	**4**	39	**3**	40	**1**
41	**4**	42	**1**	43	**4**	44	**3**	45	**4**
46	**3**	47	**12**	48	**4**	49	**36**	50	**5**

CHEMISTRY

51	**2**	52	**3**	53	**1**	54	**1**	55	**2**
56	**1**	57	**4**	58	**3**	59	**3**	60	**4**
61	**3**	62	**1**	63	**3**	64	**2**	65	**1**
66	**2**	67	**4**	68	**2**	69	**1**	70	**1**
71	**2**	72	**5**	73	**2**	74	**31**	75	**2**

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SOLUTIONS

MATHEMATICS

1. $1+10\,\mathrm{Re}\left(\dfrac{2\cos\theta+i\sin\theta}{\cos\theta-3i\sin\theta}\right)=0$

$\therefore Z+\bar{Z}=2\,\mathrm{Re}(z)$

$\dfrac{2\cos\theta+i\sin\theta}{\cos\theta-3i\sin\theta}+\dfrac{2\cos\theta-i\sin\theta}{\cos\theta+3i\sin\theta}=2\times\left(\dfrac{-1}{10}\right)$

$2\theta=\dfrac{\pi}{2},\dfrac{3\pi}{2},\dfrac{5\pi}{2},\dfrac{7\pi}{2}$

$\sum 4\theta^2=\dfrac{\pi^2}{4}+\dfrac{9\pi^2}{4}+\dfrac{25\pi^2}{4}+\dfrac{49\pi^2}{4}=\dfrac{84\pi^2}{4}=21\pi^2$

2. A, K, N, P, R,U

440th word =PRKAUN.

3. $T_{r+1}=9_{c_r}\left(x^{\frac{2}{3}}\right)^{9-r}.\left(\dfrac{x^{\frac{-2}{5}}}{2}\right)^{r}$

4. $a_2+a_{4+.....+a_n}=30$

$a_1+a_{3+.....+a_{n-1}}=24$

$A.P:\dfrac{3}{2},3,\dfrac{9}{2},6,\dfrac{15}{2},9,\dfrac{21}{2},12.$

Number of integer terms =4

5. $\left|\dfrac{Z_1}{|Z_1|}+\dfrac{Z_2}{|Z_2|}\right|\le\left|\dfrac{Z_1}{|Z_1|}\right|+\left|\dfrac{Z_2}{|Z_2|}\right|$

$\dfrac{a}{|y-z|}=\dfrac{b}{|z-x|}=\dfrac{c}{|x-y|}$

$\dfrac{a^2}{y-z}+\dfrac{b^2}{z-x}+\dfrac{c^2}{x-y}=0$

6. $\dfrac{1}{\sqrt{1}+\sqrt{2}}+\dfrac{1}{\sqrt{2}+\sqrt{3}}+...+\dfrac{1}{\sqrt{99}+\sqrt{100}}=m$

On rationalizing the denominator.

7. $y^2=12x$

$\Rightarrow a=3$

Given $SP\times SQ=\dfrac{147}{4}$

Let $P\left(3t^2,6t\right)$ and $t_1t_2=-1$ (ends of focal chord)

So, $Q\left(\dfrac{3}{t^2},\dfrac{-6}{t}\right),S(3,0).$

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$SP \times SQ = PM_1 \times QM_2$

$\beta - \alpha = 1328$

8. Conceptual.

9. Correct mean $= \mu = \dfrac{100(40) - 50 + 40}{100}$

$\mu = 40 - \dfrac{1}{10} = 39.9$

Incorrect variance: $(5.1)^2 = \dfrac{\sum xi^2}{100} - (\bar{x})^2$

$\sigma = 5$

10. Conceptual

11. Total no. of ways of distribution $= 4^{10} = 2^{20}$

Favorable outcomes $= 4_{C_2} \times 2_{C_1} \times 10_{C_2} \times 8_{C_3} \times 2^5$

$\therefore$ Required probability $= \dfrac{4_{C_2} \times 2_{C_1} \times 10_{C_2} \times 8_{C_3} \times 2^5}{2^{20}}$.

12. $D_g \in R$ and $D_f \in (0, \infty)$

For $D_{fog} \Rightarrow g(x) > 0$

$\dfrac{x^4 - 2x^3 + 3x^2 - 2x + 2}{2x^2 - 2x + 1} > 0$

Clearly $x < 0$ satisfies which are included in option (A) only

13. $\cos^{-1} x = \pi + \sin^{-1} x + \sin^{-1}(2x+1)$

$\cos^{-1} x = \pi + \dfrac{\pi}{2} - \cos^{-1} x + \sin^{-1}(2x+1)$

$2\cos^{-1} x - \sin^{-1}(2x+1) = \dfrac{3\pi}{2}$

Let $\cos^{-1} x = \alpha, \quad \sin^{-1}(2x+1) = \beta$.

14. $BCB^{-1} = A$

$(BCB^{-1})(BCB^{-1}) = A \cdot A$

$BCB^{-1}BCB^{-1} = A^2$

$BC^2B^{-1} = A^2$

$B^{-1}(BC^2B^{-1})B = B^{-1}(A^2)B$

Also $AB^{-1} = A^{-1}$

$C^2 = B$

$\alpha = -6, \beta = 2$

$\therefore (2\beta - \alpha)^2 = 100$

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15. $|A|=\begin{vmatrix}2 & 2+p & 2+p+q\\ 4 & 6+2p & 8+3p+2q\\ 6 & 12+3p & 20+6p+3q\end{vmatrix}=2^3$

$\therefore |adj(adj3A)|=|3A|^{(3-1)^2}=|3A|^4=2^{12}\times 3^{12} \Rightarrow m\times n=144$

16. $f(x+y)=f(x)f'(y)+f'(x).f(y)$

Put $x=0, y=0$ in

$f'(0)=\frac{1}{2}$

Put $y=0$ in $\Rightarrow f'(x)=\frac{f(x)}{2}$

17. Given $5f(x)+4f\left(\frac{1}{x}\right)=x^2-2 \quad \forall x\neq 0$ ……(1)

Substituting $\frac{1}{x}$ inplace of x, weget

$5f\left(\frac{1}{x}\right)+4f(x)=\frac{1}{x^2}-2$ …… (2)

On solving 1&2 weget

$f(x)=\frac{5x^4-2x^2-4}{9x^2}$

For strictly increasing $f'(x)>0$

18. Let $I=\int\left(\frac{1}{x^2}+\frac{1}{x^4}\right)\left(\frac{3}{x}+\frac{1}{x^3}\right)^{\frac{1}{23}}dx$

Put $\frac{3}{x}+\frac{1}{x^3}=t$

$\alpha+\beta+\gamma=19$

19. $10\left(\sin^2\theta\right)^2+15\left(1-\sin^2\theta\right)^2=6$

$\sin^2\theta=\frac{3}{5}$

20. Let $y=\frac{5-x}{x^2-3x+2}$

If $y=0$ then $x=5$

If $y\neq 0\Rightarrow D\geq 0$

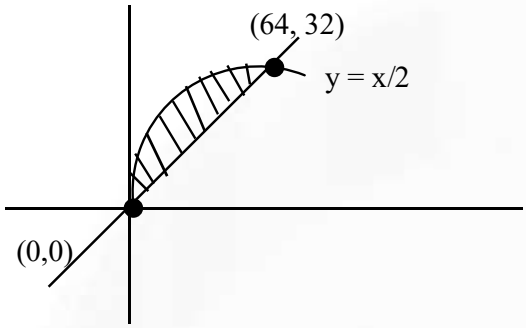
21. $\int_0^{\frac{\pi}{4}} e^{-x}.\tan_x^{50}dx=\left[-e^{-x}\tan_x^{50}\right]_0^{\frac{\pi}{4}}+\int_0^{\frac{\pi}{4}} e^{-x}.50.\tan_x^{49}.\sec_x^2 dx$

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$$= -e^{\frac{-\pi}{4}} + 50\left[\int_0^{\frac{\pi}{4}} e^{-x}\left[\tan_x^{51} + \tan_x^{49}\right]\right]dx$$

Given integral = 50

22.



$|x-y| \le y \le 4\sqrt{x}$

Now $y = |x-y|$

$\Rightarrow y^2 = (x-y)^2$

$\Rightarrow y = \frac{x}{2}$ and x = 0

Area $= \int_0^{64}\left(4\sqrt{x} - \frac{x}{2}\right)dx$

23. $\left(x^2+1\right)\frac{dy}{dx} - 2xy = \left(x^4+2x^2+1\right)\cos x$

$= \left(x^2+1\right)^2 \cos x$

$\frac{dy}{dx} - \left(\frac{2d}{x^2+1}\right)y = \left(x^2+1\right)\cos x$

$y = \left(x^2+1\right)(1+\sin x)$

$\therefore \int_{-3}^{3} y\,dx = 24$

24. Conceptual

25. S. D. $= \frac{\left|\left[\bar{a}-\bar{c} \quad \bar{b} \quad \bar{d}\right]\right|}{\left|\bar{b}\times\bar{d}\right|}$

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PHYSICS

26. Conceptual

27. Folding method

B, D have same potentials

$\therefore$ Two 5N Resistors are motive

$\text{Req} = \frac{8}{5} \quad \therefore i = \frac{v}{R} = \frac{16}{8} \times 5 = 10^{A}$

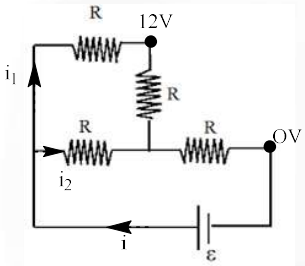
Current Division Rule:

$I_{Ac} = 10\left(\frac{2}{10}\right) = 2A$

28. Conceptual

29. In steady stat capacitor is open

$\therefore \text{Re}q = \frac{5R}{3} \quad \therefore i = i_1 + i_2$



$i = \frac{E}{R_e} = \frac{3E}{5R}$

$i_1 = \frac{R}{3R} \cdot \frac{3E}{5R}$

$i_1 = \frac{3E}{15R} = \frac{E}{5R}$

KVL: $12 - i_1 R - iR = 0$

$12 - \frac{E}{5R}.R - \frac{3E}{5R}.R = 0$

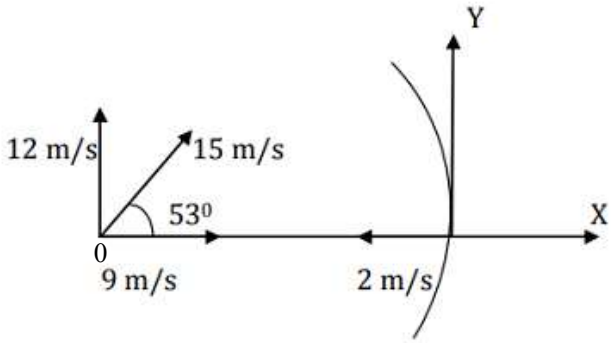
$12 = \frac{E}{5} + \frac{3E}{5} = \frac{4}{5}E$

$\therefore E = \frac{12 \times 5}{4} = 15V$

30. Conceptual

31. Conceptual

32.



$\vec{V}_0$=Velocity of object $= \left(9\hat{i} + 2\hat{j}\right) m/s$

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$\vec{V}_m$ = Velocity of mirror = $-2\hat{i}m/s$

$$m=\frac{f}{f-u}=\frac{-20}{-20-(-30)}=-2$$

$$\left(\vec{V}_{1/m}\right)=-m^2\left(\vec{V}_{0/m}\right)$$

$$\left(\vec{V}_{1/m}\right)=(-2)^2 11\hat{i}=-44\hat{i}m/s$$

For velocity component perpendicular to optical axis

$$\left(\vec{V}_{1/m}\right)=\left(\vec{V}_{o/m}\right)=(-2)12\,\hat{j}=-24\,jm/s$$

$\therefore \vec{V}_{1/m}$ = Velocity of image w.r.t. mirror

$=\left(\vec{V}_{1/m}\right)+\left(\vec{V}_{1/m}\right)=\left(-44i-24\,\hat{j}\right)m/s$ Also, $\vec{V}_{1/m}=\vec{V}_1-\vec{V}_m$

Or $\vec{V}_1=\left(-44\hat{i}-24\,\hat{j}\right)-2\hat{i}=\left(-46\hat{i}-24\,\hat{j}\right)m/s$

33. Initially the flux linked with each turn of coli is given as

$$\phi=\vec{B}.\vec{A}=BA\cos\theta=BA\cos 45^0$$

Substituting the values, we get

$$\phi=2.0\times300\times10^4\times0.7071$$

$$\Rightarrow\phi=4.24\times10^{-2}Wb$$

Finally flux linked with each turn of the coli is given as

$$\phi'=BA\cos 0^0=BA$$

$$\Rightarrow\phi'=2.0\times300\times10^{-4}=6.0\times10^{-2}Wb$$

Change in flux in this process is given as

$$\Delta\phi=\phi'-\phi$$

$$\Rightarrow\Delta\phi=\left(6.0\times10^{-2}\right)-4.24\times10^{-2}$$

$$\Rightarrow\Delta\phi=1.76\times10^{-2}Wb$$

This change is carried out in $0.1s$. The magnitude of the average EMF induced in the coli is given as

$$e=N\frac{d\left(\phi'-\phi\right)}{dt}=50\times\frac{1.76\times10^{-5}}{0.1}$$

$$\Rightarrow\Rightarrow e=8.8V$$

34. $$\frac{d\theta}{dt}=\frac{\sigma eA}{ms}\left(T^4-T^4\right)$$

$$\frac{\left(\frac{d\theta}{dt}\right)_1}{\left(\frac{d\theta}{dt}\right)_2}=\frac{A_1}{A_2}\times\frac{m_2}{m_1}\times\frac{\left[(400)^4-(300)^4\right]}{\left[(600)^4-(300)^4\right]}$$

$$=\frac{4\pi R^2}{4\times4\pi R^2}\times\frac{\rho\times4/3\pi\times8R^3}{\rho\times4/3\pi R^3}$$

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$$\frac{\left(\frac{d\theta}{dt}\right)_1}{\left(\frac{d\theta}{dt}\right)_2}=2\times\left(\frac{256-81}{1296-81}\right)=\frac{2\times(175)}{1215}$$

$$\left(\frac{d\theta}{dt}\right)_2=\frac{\left(\frac{d\theta}{dt}\right)_1\times 1215}{175\times 2}$$

$$=\frac{2.8\times 1215}{175\times 2}=9.72^0 C/\min$$

35. Let x_0 be the extension in the spring in equilibrium. Then, equilibrium of A and B give,

$$T=kx_0+mg\sin\theta------(1)$$

$$2T=mg------(2)$$

Here, T is the tension in the string. Now, suppose A is further displaced by a distance x from its mean position and υ be its speed at this moment. Then, B lower by $\frac{x}{2}$ and speed of B at this instant will be $\frac{\upsilon}{2}$. Total energy of the system in this position will be,

$$E=\frac{1}{2}k(x+x_0)^2+\frac{1}{2}m_A\upsilon^2+\frac{1}{2}m_B\left(\frac{\upsilon}{2}\right)^2+m_Agh_A-m_Bgh_B$$

$$E=\frac{1}{2}k(x+x_0)^2+\frac{1}{2}m\upsilon^2+\frac{1}{8}m\upsilon^2+mgx\sin\theta-mg\frac{x}{2}$$

$$E=\frac{1}{2}k(x+x_0)^2+\frac{5}{8}m\upsilon^2+mgx\sin\theta-mg\frac{x}{2}$$

Since, E is constant, $\frac{dE}{dt}=0$

$$0=k(x+x_0)=\frac{dx}{dt}+\frac{5}{4}m\upsilon\left(\frac{d\upsilon}{dt}\right)+mg(\sin\theta)\left(\frac{dx}{dt}\right)-\frac{mg}{2}\left(\frac{dx}{dt}\right)$$

Substituting, $\frac{dx}{dt}=\upsilon\Rightarrow\frac{d\upsilon}{dt}=a$

And $kx_0+mg\sin\theta=\frac{mg}{2}$

We get, $\frac{5}{4}ma=-kx$

Since, $a\alpha-x$

Motion is simple harmonic, time period of which is,

$$T=2\pi\sqrt{\left|\frac{x}{a}\right|}=2\pi\sqrt{\frac{5m}{4k}}\qquad\omega=\frac{2\pi}{T}=\sqrt{\frac{4k}{5m}}$$

36. $PV=RT;\left(P_0-\alpha V^2\right)V=RT$

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For maximum temperature $\frac{dT}{dV}=0$

$$\frac{dT}{dV}=\left(\frac{P_0}{R}-\frac{\alpha}{R}\times 3V^2\right)=0\Rightarrow V\left(\frac{P_0}{3\alpha}\right)^{\frac{1}{2}}$$

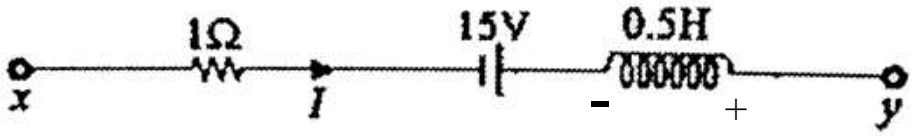
T is maximum at $V=\left(\frac{P_0}{3\alpha}\right)^{\frac{1}{2}}$

$$\therefore\left(P_0-\alpha\times\frac{P_0}{3\alpha}\right)\left(\frac{P_0}{3\alpha}\right)^{\frac{1}{2}}=RT_{\max}\Rightarrow T_{\max}=\frac{2P_0}{3R}\times\left(\frac{P_0}{3\alpha}\right)^{\frac{1}{2}}$$

37. Conceptual

38. In the above circuit section as current is decreasing with time, the polarity of self induced EMF is shown in figure-5.101. In this branch of circuit we equation of potential drop from point x to y which is given as

$$V_x-iR+E+L\frac{di}{dt}=V_y\Rightarrow V_x-5\times 1+15+0.5\times 10=V_y \qquad \Rightarrow V_y-V_x=15V$$



39. The experimental setup is in a liquid, therefore the wavelength of light will change.

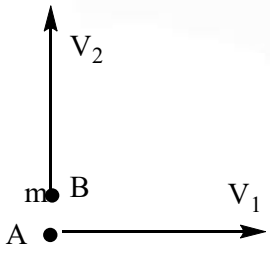
$$\lambda_{liquid}=\frac{\lambda_{air}}{\mu}=\frac{6300}{1.33}=\frac{6300\times 10^{-10}}{1.33}m$$

Fringe width $\beta=\frac{\lambda_{liquid}}{d}=\frac{\lambda_{air}D}{\mu d}$ $=\frac{6300\times 10^{-10}}{1.33}\times\frac{1.33}{10^{-3}}=6.3\times 10^{-4}m$

40. $I_d=\varepsilon_0\frac{d\phi}{dt}=\varepsilon_0 A\frac{dE}{dt}=\varepsilon_0\frac{A}{\ell}\frac{dV}{dt}$

$$I_d=\frac{\varepsilon_0\times\pi\times 4\times 10^{-4}\times 5\times 10^{13}}{10^{-1}}=\frac{5\times 10^{14-4}}{9\times 10^9}=5.5A$$

41. $v_{cm}=\frac{m_1v_1+mv_2}{m+m}=\frac{v_1+v_2}{2}$



$$v_{1cm}=v_1-v_{cm}=v_1-\frac{v_1+v_2}{2}=\frac{v_1-v_2}{2}$$

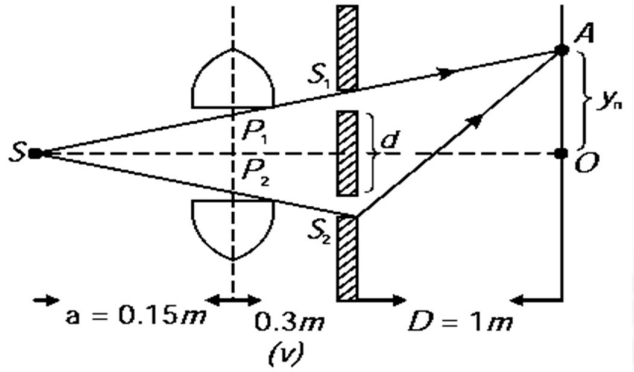
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$$\left|v_{1,cm}\right| = \frac{1}{2}\sqrt{v_1^2 + v_2^2 - 2v_1v_2\cos 90^o} = \frac{1}{2}\sqrt{v_1^2 + v_2^2}$$

So, required wavelength is

$$\lambda = \frac{h}{m\left|v_{1,cm}\right|} = \frac{2h}{\sqrt{v_1^2 + v_2^2}} = \frac{2h}{\sqrt{\left(\frac{h}{\lambda_1}\right)^2 + \left(\frac{h}{\lambda_2}\right)^2}} = \frac{2\lambda_1\lambda_2}{\sqrt{\lambda_1^2 + \lambda_2^2}}$$

42.



The image of S due to both parts of lenses are formed at S_1 and S_2 whose distance from the lens is given by $\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \Rightarrow \frac{1}{0.1} = \frac{1}{v} - \frac{1}{(0.15)}$

$$\frac{1}{v} = \frac{1}{0.1} - \frac{1}{0.15} = \frac{10}{1} - \frac{100}{15} = \frac{150-100}{15} = \frac{50}{15} = \frac{10}{3}$$

$$v = \frac{3}{10} = 0.3m$$

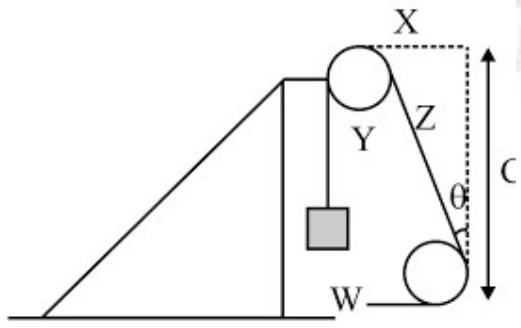
from similar Δss_1s_2 and sp_1p_2 $\qquad \frac{d}{0.5\times10^{-3}} = \frac{0.3+0.15}{0.15}$

$$d = \frac{0.45}{0.15}\times 0.5\times10^{-3} = 1.5\times10^{-3}m$$

For third maxima $y_n = \frac{3\lambda D}{d} = \frac{3\times500\times10^{-9}\times1}{1.5\times10^{-3}} = 10^{-9}\times10^6$

$$y_n = 10^{-3}m \Rightarrow y_s = 1mm$$

43. $z = \sqrt{x^2 + c^2}$



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Now, $w+y+z=l$

or $w+y+\sqrt{x^2+c^2}=l$

$\therefore \quad \frac{dw}{dt}+\frac{dy}{dt}+\frac{x}{\sqrt{x^2+c^2}}\cdot\frac{dx}{dt}=0$

or $\left(-\frac{dw}{dt}\right)+\frac{x}{z}\left(-\frac{dx}{dt}\right)=\frac{dy}{dt} \quad ....(i)$

$-\frac{dw}{dt}=-\frac{dx}{dt}=v_2$

$\frac{dy}{dt}=v_1$

and $\frac{x}{z}=\sin\theta$

Substituting these values in Eq. (i) we have

$\frac{dy}{dt}=v_1$

$v_2(1+\sin\theta)=v_1$

44. $Y=9\left(A\cdot(A\cdot B)\cdot\left((B\cdot(A\cdot B)=A\cdot(A\cdot B)+B\cdot(A\cdot B)\right)\right.$

$=A\cdot\left(\overline{A}+\overline{B}\right)+B\cdot\left(\overline{A}+\overline{B}\right)$

$=A\cdot\overline{A}+A\cdot\overline{B}+\overline{A}\cdot B+B\cdot\overline{B}$

$=A\cdot\overline{B}+\overline{A}\cdot B$

When $A=1,B=0$ then $Y=1.\overline{0}+\overline{1}.0=1$

When $A=1,B=1$ then $Y=1.\overline{1}+\overline{1}.1=0$

When $A=0,B=0$ then $Y=0.\overline{0}+\overline{0}.0=0$

45. Let a = vertically downward acceleration of COM and $\alpha=$ angular acceleration of the stick

Using $F=Ma_{CM}$

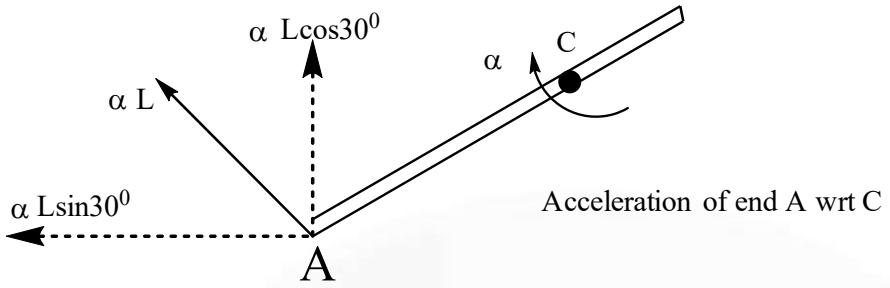
$Mg-T=Ma \quad ....(i)$

Using $\tau=I\alpha$ about COM

$T.L\cos30^\circ=\frac{M(2L)^2}{12}\cdot\alpha$

$\Rightarrow \quad T=\frac{2ML\alpha}{3\sqrt{3}} \quad ....(ii)$

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Acceleration of end A relative to COM is αL in a direction perpendicular to rod. Vertical component of this acceleration I s $\alpha L\cos 30^0$

For A to have no acceleration in ground frame

$$\alpha L\cos 30^0 = a \Rightarrow \frac{\sqrt{3}}{2}\alpha L = a \ldots\ldots\ldots\ldots \text{(iii)}$$

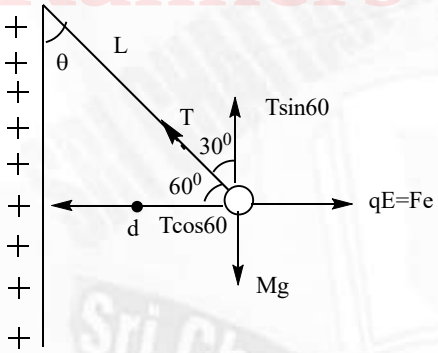
Putting this in (ii) gives $T = \frac{4Ma}{g} \ldots\ldots\ldots$ (iv)

Put this in (i) $Mg - \frac{4ma}{9} = Ma \quad \Rightarrow \quad a = \frac{9g}{13}$

$$T = \frac{4M}{9}.\frac{9g}{13} = \frac{4mg}{13}$$

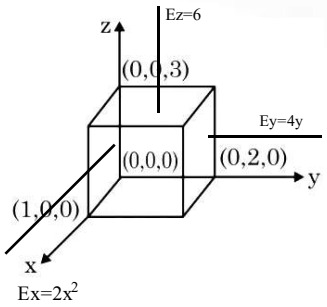
46. $\sin\theta = \frac{d}{L} = \frac{10\times10^{-2}}{20\times10^{-2}} = \frac{1}{2}$

$\theta = 30^0$



$$\tan 60 = \frac{mg}{qE} \therefore q = \frac{0.2}{2\times10^4\times\sqrt{3}} \quad q = \frac{1}{\sqrt{3}}\mu c$$

47. $\phi = \bar{E}.\bar{A}$



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$\phi_{0x} = 2\times 6 = 12\dfrac{wb}{m^2}$

$\phi_{0y} = -3\times 4\times 2 = -24$

$z = 0, z = 3$

$\phi_{oz} = (6\times 2) + (6\times 2) = 24$

$\phi_{net} = 12 = \dfrac{q}{z}$

$q = 12z$

$\therefore n = 12$

48. The situation is shown in figure -5.90. If mass of disc is m, then mass of rod $= \eta m$, If v_1 and v_2 be the velocity of the disc and rod after collision, using linear momentum conservation, we have

$mv = mv_1 + \eta m v_2$

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Or $v = v_1 + \eta v_2$

If after collision, rod starts rotating with angular speed ω, using conservation of angular momentum, we have $mv\dfrac{L}{2} = \left(\dfrac{\eta mL^2}{12}\right)\omega + mv_1\dfrac{L}{2}$

Or $v = \dfrac{1}{6}\eta L\omega + v_1$

Or $\omega = \dfrac{6(v - v_1)}{\eta L}$

As collision is elastic, using kinetic energy conservation, we get

$\dfrac{1}{2}mv^2 = \dfrac{1}{2}mv_1^2 + \dfrac{1}{2}\eta m v_2^2 + \dfrac{1}{2}I\omega^2$

Or $v^2 = v_1^2 + \dfrac{3}{\eta}(v - v_1)^2 + \eta\dfrac{(v - v_1)^2}{\eta^2}$

Solving for v_1, we get $v_1 = \dfrac{4-\eta}{4+\eta}v$ $\quad\therefore\quad \eta = 4; v_1 = 0$

49. Given data,

$L = 1m, r = 10^{-3}, \quad e = 1, \quad T = 900k,$

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$T_0 = 300k, \quad \rho_{300k} = \pi^2 \times 10^{-8}\Omega - m \qquad \alpha = 7.8\times10^{-3}/^0 C, I = ?$

Heat energy generated per sec in wire = Energy radiated by wire per sec

According to given problem wire will behave like black body so $e = 1$ and at equilibrium condition temperature of the wire will be 900K so,

$$I^2 R_{900} = \sigma Ae\left[T^4 - T_0^4\right] \; I = \left[\frac{\sigma Ae\left[T^4 - T_0^4\right]}{R_{900}}\right]^{\frac{1}{2}} \; I = \left[\frac{\sigma Ae\left[T^4 - T_0^4\right]}{\frac{\rho_{900}}{a}^{L}}\right]^{\frac{1}{2}}$$

Here L=length, a = wire cross-section area

$$A = 2\pi rL, \quad a = \pi r^2, \quad e = 1$$

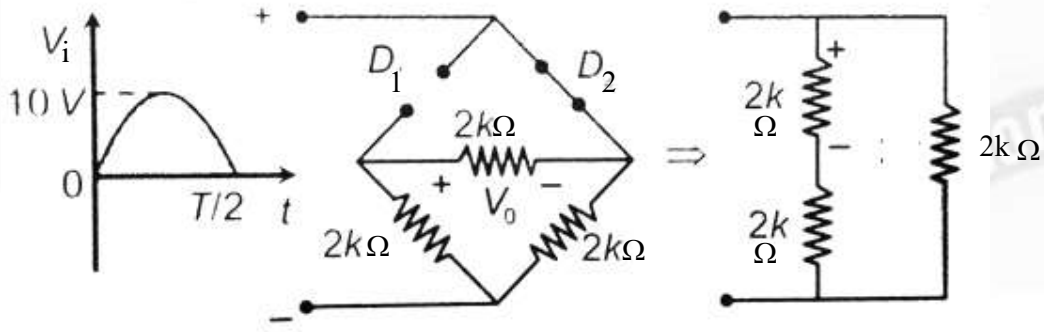
$$I = \left[\frac{\sigma \times 2\pi rL \times \left(T^4 - T_0^4\right) \times a}{\rho_{300}\left(1+\alpha 600\right) \times L}\right]^{\frac{1}{2}} = \left[\frac{\sigma \times 2\pi^2 r^3 \times \left(T^4 - T_0^4\right)}{\pi^2 \times 10^{-8} \times \left(1 \times 7.8 \times 10^{-3} \times 600\right)}\right]^{\frac{1}{2}}$$

$$I = \left[\frac{5.68\times10^{-8} \times 2 \times \pi^2 \times 10^{-9} \times \left[(900)^4 - (300)^4\right]}{\pi^2 \times 10^{-8} \times \left(1 + 7.8 \times 600 \times 10^{-3}\right)}\right]^{\frac{1}{2}}$$

$$I = \left[\frac{5.68 \times 2 \times 10^{-9} \times 10^8 \left[(9)^4 - (3)^4\right]}{\left(1 + 7.8 \times 600 \times 10^{-3}\right)}\right]^{\frac{1}{2}} = \left[\frac{5.68 \times 0.2 \times [6561 - 81]}{(5.68)}\right]^{\frac{1}{2}}$$

$$I = [1296]^{\frac{1}{2}} = 36\, amp$$

50. For the positive half cycle of input the resulting network is shown below



$$\Rightarrow (V_0)_{\max} = \frac{1}{2}(V_i)_{\max} = \frac{1}{2} \times 10 = 5v$$

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CHEMISTRY

51. RLVP is equal to the mole fraction of solute but not solvent

52. $NaCl + AgNO_3 \rightarrow \underset{(A)}{AgCl}(\downarrow) + \underset{(B)}{NaNO_3}$

$AgCl + 2NH_3 \rightarrow \left[Ag(NH_3)_2\right]Cl$

$CuSO_4 + 4NH_3 \rightarrow \left[Cu(NH_3)_4\right]SO_4$

By adding HNO_3 reaction will be shifted in backward Direction to give more $AgCl$

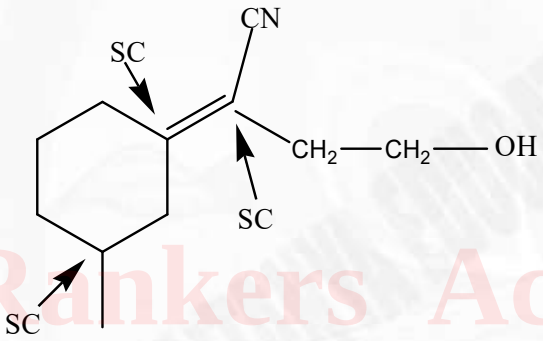
In complex D, $n = 1$, $\mu = \sqrt{n(n+2)}BM$

$= \sqrt{1(1+2)}BM$

$= 1.732$ BM

53. ΔiH order: $Li < Be > B < C < N > O < F$

54.

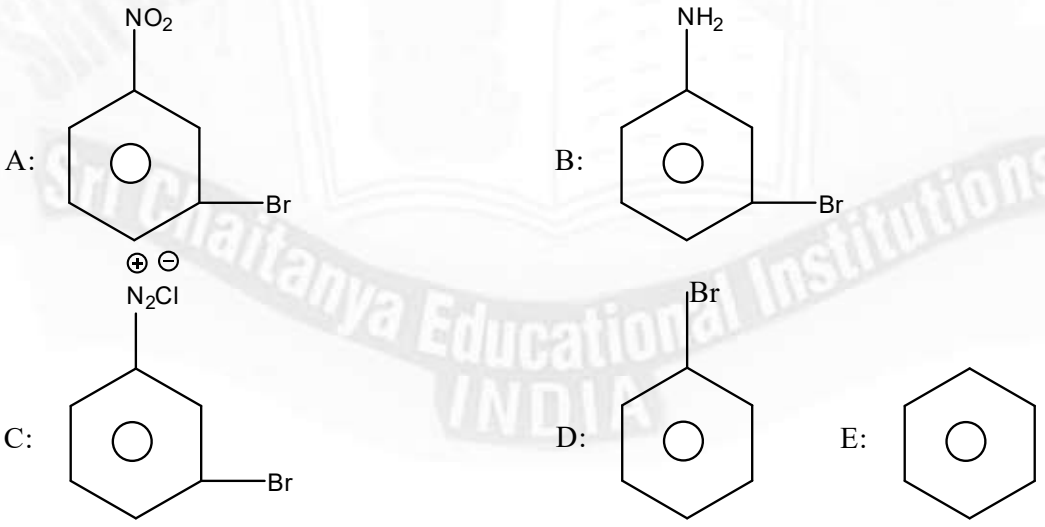


SC → Stereogenic centre

It has 3 stereo centres

It shows both geometrical and optical Isomerism

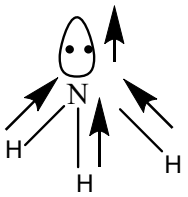
55.

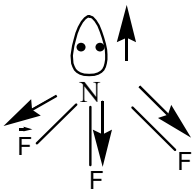


56. As per MOT, C_2 gas molecule contains two pi bonds due to the presence of four electrons in two pi molecular orbitals.

NH_3 has bond pair & lone pair moments in the same directions but in NF_3 they are opposite.

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$\mu = 4.9\times10^{-30} cm$ $\mu = 0.8\times10^{-30} cm$

57. $C_6H_6(\ell)+\frac{15}{2}O_2(g)\rightarrow 6CO_2(g)+3H_2O(\ell)$

$\Delta_C H^\Theta = \Delta_r H^\Theta$

$$-3267=\left[6\Delta_f H^\Theta(CO_2)+3\Delta_f H^\Theta(H_2O)\right]-\left[\Delta_f H^\Theta(C_6H_6)+\frac{15}{2}\Delta_f H^\Theta(O_2)\right]$$

$$-3267=\left[6(-393.5)+3(-285.83)\right]-\left[\Delta_f H^\Theta_{(C_6H_6)}+\frac{15}{2}\times 0\right]$$

$$\Delta_f H^\Theta_{(C_6H_6)}=\left[-2361-857.49\right]+3267$$

$$=48.51 kJmol^{-1}$$

58. Number of spectral lines $=\frac{n(n-1)}{2}=\frac{4(4-1)}{2}=6$

Number of radial nodes $=n-l-1=4-0-1=3$

Number of degenerate orbitals in M-shell of H atom

`

$$=1(3s)+3(3p)+5(3d)$$

$$=9$$

$Na: 1s^1 2s^2 2p^6 3s^1 \rightarrow n=3, l=0, m=0, s=\frac{1}{2}$

59. $A + B_2 \rightarrow AB_2$

Initial $\rightarrow$ 300 200 0

Final $\rightarrow$ 100 0 200

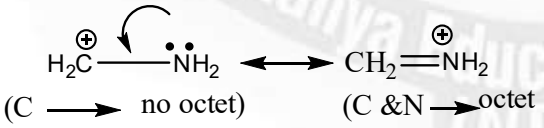
60. Atomic radius: $B < Ga < Al < In < T\ell$

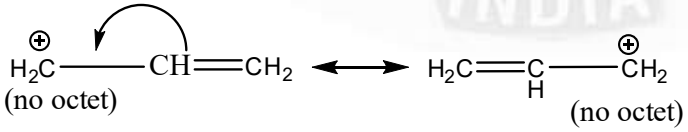
I^{st} Ionisation enthalpy: $In < Al < Ga < T\ell < B$

Electronegativity: $Al < Ga < In < T\ell < B$

Melting point: $Ga < In < T\ell < Al < B$

61.





62. $2I^- + S_2O_8^{2-} \xrightarrow{Fe^{3+}} I_2 + 2SO_4^{2-}$

Mechanism

$2Fe^{3+} + 2I^- \rightarrow 2Fe^{2+} + I_2$

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$2Fe^{2+} + S_2O_8^{2-} \rightarrow 2Fe^{3+} + 2SO_4^{2-}$

63. $HCl + H_2S$ is the group –II reagent it causes precipitation of group –I & II Cations but not group-II, IV, V and VI.

Group- I: Ag^+

Group-II: Cu^{2+}

Group-III: Al^{3+}, Fe^{3+}

Group-IV: Zn^{2+}, Ni^{2+}

Group-V: Ba^{2+}, Ca^{2+}

64. In the denaturation of protein, its primary structure is not destroyed.

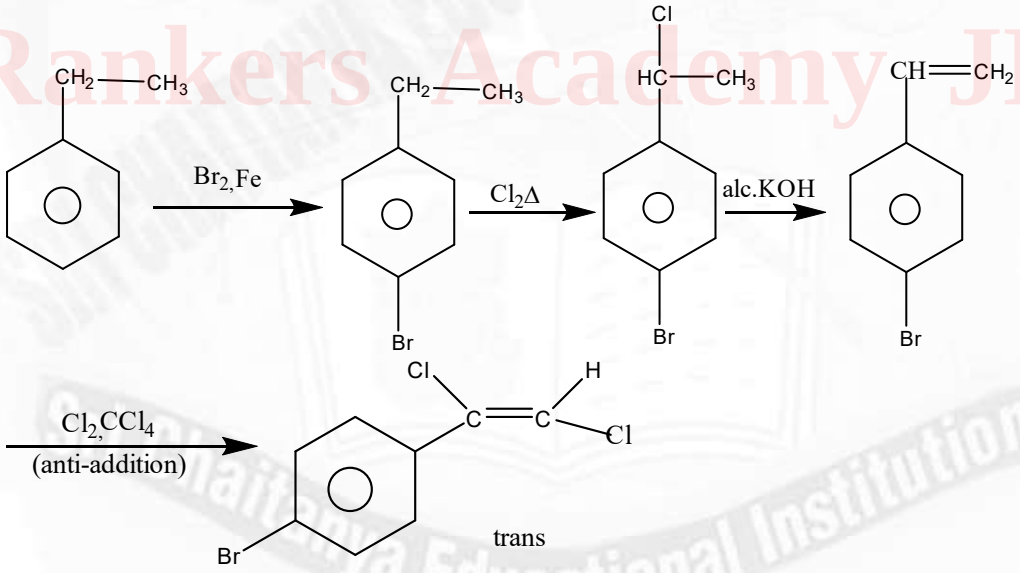
65. $Kt = 2.303\log\left(\frac{a}{a-x}\right)$

$Kt_1 = 2.303\log\frac{a}{a/2}$

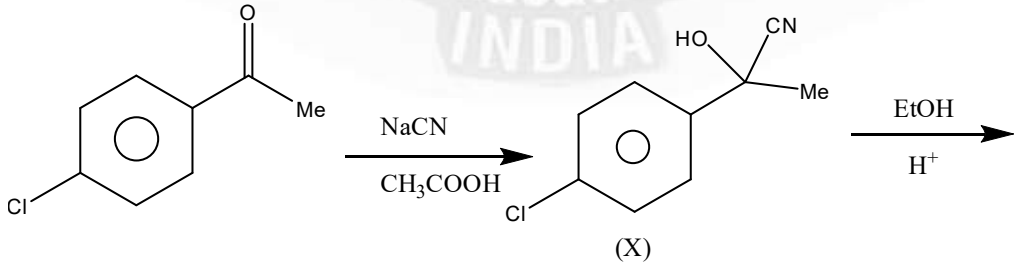
$Kt_2 = 2.303\log\frac{a}{\left(\frac{a}{4}\right)}$

$\frac{t_1}{t_2} = \frac{\log 2}{\log 4} = \frac{\log 2}{2\log 2} = \frac{1}{2}$

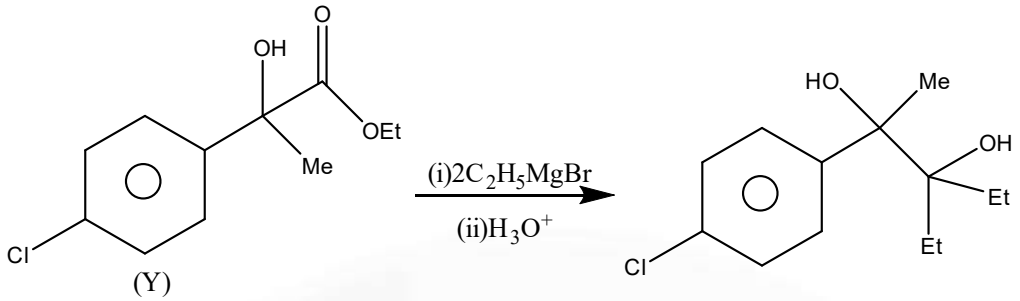
66.



67.



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68. $-ve \quad \Delta egH \quad order : Cl > F > Br > I$

$: S > Se > Te > P_o > O$

69. For the salt of weak acid and weak base,

$$P^H = \frac{1}{2}\left[P^{kw} + P^{ka} - P^{kb}\right]$$

$$= \frac{1}{2}[14 + 4.76 - 4.75]$$

$$= 7.005$$

70. $$\Delta G^{\Theta} = -nFE^{\Theta}_{cell}$$

$$= -2 \times 96500 \times 1.1$$

$$= 212300\, J / mol$$

$$= -212.3\, kJ / mol$$

71. $Cathode : Cu^{2+}_{aq} + 2e^- \rightarrow Cu_{(s)}$

$anode : Mn_{(s)} \rightarrow Mn^{2+}_{aq} + 2e^-$

$cell\ reaction : Cu^{2+}_{(aq)} + Mn_{(s)} \rightarrow Mn^{2+}_{(aq)} + Cu_{(s)}$

$$Q_c = \frac{\left[Mn^{2+}\right]}{\left[Cu^{2+}\right]}$$

$$Mn(OH)_{(s)} \rightleftharpoons Mn^{2+}_{(aq)} + 2OH^-_{(aq)}$$

$$K_{sp} = \left[Mn^{2+}\right]\left[OH^-\right]^2$$

$$1.9 \times 10^{-13} = \left[Mn^{2+}\right] \times \left(10^{-4}\right)^2$$

$$\left[Mn^{2+}\right] = \frac{1.9 \times 10^{-13}}{10^{-8}} = 1.9 \times 10^{-5} m$$

$$E_{cell} = E^{\Theta}_{cell} - \frac{0.059}{n}\log Q_c$$

$$= (0.34 + 1.18) - \frac{0.059}{2}\log\left(\frac{1.9 \times 10^{-5}}{0.675}\right)$$

$$= 1.52 - 0.295 \times \log\left(2.81 \times 10^{-5}\right)$$

$$= 1.52 - 0.295(-5 + 0.45)$$

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$=1.52+0.137$

$=1.657V\sim 2V$

72. $O_2, \bar{C}_2, \bar{O}_2, H_2^+$ and He_2^+ are paramagnetic due to the presence of unpaired electrons in the molecular orbitals.

73. 

During acetylation, one H is lost and one $COCH_3$ group is added. So, the net increase in molecular mass is 42 per one $-NH_2$ group

No .of NH_2 groups $=\dfrac{192-108}{42}=\dfrac{84}{42}=2$

74. $X+\quad Y \rightleftharpoons 2Z$

$Ci \rightarrow 1.5 \quad 0.5$

$Ceq \rightarrow 1-x \quad 1.5-x \quad 0.5+2x$

Given, $0.5+2x=1 \Rightarrow x=0.25M$

$K_c=\dfrac{[Z]^2}{[X][Y]}$

$=\dfrac{1^2}{0.75\times 1.25}=\dfrac{16}{15}=\dfrac{x}{y}$

$x+y=31$

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75. $Fe^{2+}\rightarrow Fe^{3+}+e^-\left(n_f=1\right)$

$C_2O_4^{2-}\rightarrow 2CO_2+2e^-\left(n_f=2\right)$

$MnO_4^-+5e^-\rightarrow Mn^{2+}\left(in\,acidic\,medion\right)\left(n_f-5\right)$

eq. of Oxidant =eq. of reductant

a) $5\times n_{KMnO_4}=3\times 1\left(FeC_2O_4\right)$

n_{KMnO_4} for $FeC_2O_4=\dfrac{3}{5}$

b) $5\times n_{KMnO_4}=6\times 1\left(Fe_2\left(C_2O_4\right)_3\right)$

n_{KMnO_4} for $Fe_2\left(C_2O_4\right)_3=\dfrac{6}{5}$

c) $5\times n_{KMnO_4}=1\times 1\left(FeSO_4\right)$

n_{KMnO_4} for $FeSO_4=\dfrac{1}{5}$

d) $Fe_2\left(SO_4\right)_3$ is not oxidised

$\therefore n_{KMnO_4}\left(total\right)=\dfrac{3}{5}+\dfrac{6}{5}+\dfrac{1}{5}=2$

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